Computational Prediction of Antimicrobial Resistance in Klebsiella pneumoniae

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ABSTRACT

Antimicrobial resistance (AMR) in *Klebsiella pneumoniae* poses a critical global health threat, especially with the emergence of carbapenem-resistant strains. This study used computational methods to predict antibiotic resistance patterns by analyzing 4,003 *K. pneumoniae* isolates from the BV-BRC database. Resistance genes were identified using ResFinder, and machine learning models were created to predict resistance across 20 antibiotics. SHAP (SHapley Additive exPlantations) analysis was used to determine which genes most strongly influenced predictions. The analysis showed high resistance rates to beta-lactam antibiotics, with over 80% resistance to ampicillin. Efflux pump genes oqxA and oqxB were present in over 90% of isolates, while beta-lactamase genes like blaCTX-M-15 and blaTEM-1B were frequently detected. Machine learning models demonstrated substantially improved accuracy compared to rule based prediction where XGBoost showed up to 21.6% point improvements for certain antibiotics. SHAP analysis identified aminoglycoside-modifying enzymes (armA, aac(6')-Ib) as main predictors for amikacin resistance and carbapenemase genes (blaKPC, blaNDM, blaOXA-48) as more dominant predictors for carbapenem resistance. Overall, these findings indicate that computational approaches using machine learning can predict antimicrobial resistance patterns more accurately than gene presence alone, but specifically limitations indicate that these tools should complement rather than replace traditional phenotypic testing.

INTRODUCTION

Background on Antimicrobial Resistance

Antimicrobial resistance (AMR) has become one of the biggest health problems facing the world today. When antibiotics such as penicillin were first discovered in the early 1900s, scientists thought they had finally found a way to overcome bacterial infections. However, it turned out that bacteria are incredibly good at finding ways to survive antibiotic treatments, including beta-lactams, fluoroquinolones, aminoglycosides, and even last-resort carbapenems (1). This natural ability to evolve resistance, combined with doctors prescribing antibiotics too often, has created a serious crisis where many bacterial infections are becoming much harder to treat (1).

Klebsiella pneumoniae as a Critical Pathogen

Klebsiella pneumoniae presents one of the most significant AMR challenges. This bacterium used to mainly cause infections in people who were immunocompromised, had underlying medical conditions, or were hospitalized for other ailments, but now it has become a "superbug" that can cause serious infections in healthy people too. What makes *K. pneumoniae* so dangerous is that it can quickly develop resistance to multiple antibiotics, making it harder to eliminate (2). The bacteria have several mechanisms to fight off antibiotics, including breaking down the drugs with special enzymes called beta-lactamases (β-lactamases), changing the targets that antibiotics try to attack, pumping antibiotics out of their cells, and making their cell walls harder for drugs to get through (3), (4). These resistance traits are often carried on pieces of DNA called plasmids that can be easily shared between different bacteria, which helps resistance spread quickly through bacterial populations (2).

It is important to distinguish between intrinsic and acquired resistance in *K. pneumoniae.* Intrinsic resistance refers to inherent characteristics that make the bacteria naturally resistant to certain antibiotics, such as chromosally encoded genes that are constitutively expressed in all members of the species. For example, *K. pneumoniae* shows intrinsic resistance to ampicillin due to the chromosomal SHV-1 β-lactamase gene. In contrast, acquired resistance occurs when bacteria gain new resistance mechanisms through mutations or horizontal gene transfer of mobile genetic elements like plasmids and transposons. Understanding this distinction is crucial for predicting resistance patterns, since acquired resistance can spread rapidly between bacterial strains and even across different species, while intrinsic resistance is stable and predictable within a species.

Limitations of Current Testing Methods

The problem has gotten worse because some strains of *K. pneumoniae* have become resistant to carbapenems, which are antibiotics that doctors use as a last resort when other treatments do not work. These bacteria are especially troublesome in hospitals because they can survive on surfaces and medical equipment for long periods and form protective films called biofilms that make them even harder to kill (5).

The problem is exacerbated by the 24–48 hour time it takes to determine which antibiotic is effective for the bacterial infection, during which critically ill patients cannot wait for results. During this waiting period, doctors have to make empirical treatment decisions about which antibiotics to use, and they often choose the wrong ones or use antibiotics that are not necessary. This guessing game not only puts patients at risk but also creates more resistance by giving bacteria more chances to adapt to different drugs (Harbarth & Samore, 2005). Because current testing methods are too slow, scientists have been working on computer-based ways to predict which antibiotics will work against bacteria like *K. pneumoniae*. These computational methods could potentially solve the timing problem by quickly analyzing bacterial DNA or other characteristics to predict resistance patterns. Some researchers suggest that advanced laboratory systems using molecular techniques might be able to detect resistant bacteria in just 30 minutes (6). This kind of speed could completely change how doctors treat infections by giving them the information they need to choose the right antibiotics from the start instead of having to make empirical guesses.

Computational Approaches for Predicting Resistance

There are several different computational approaches that researchers are exploring. Some methods, such as whole-genome sequencing and bioinformatics analysis, analyze the known genetic sequences of bacteria to look for genes that cause resistance, while others use machine learning algorithms that can learn patterns from large datasets of bacterial infections and their treatment outcomes (7), (8). Prediction of antibiotic resistance in Escherichia coli from large-scale pan-genome data. PLOS Computational Biology, 14(12), e1006258. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5889021/)).Some computational tools, including phylogenetic analysis and evolutionary modeling, focus on understanding how bacteria evolve resistance over time, which could help predict what kinds of resistance might develop in the future (9). These approaches all have the potential to not only speed up resistance detection but also help doctors understand why certain treatments fail and how to choose better alternatives.

Challenges in Clinical Implementation

However, moving these computational tools from research labs into actual hospitals and clinics is not simple. Many of the current methods only work well for specific types of resistance or have only been tested in limited situations, so it is not clear if they would work reliably in real-world medical settings (10). Also, bacterial resistance is complicated because the same resistance genes might work differently in different environments, and bacteria in actual patient infections behave differently than bacteria grown in laboratory dishes (3), (11). All of these factors make it challenging to create computational models that are accurate enough for doctors to trust when making treatment decisions.

Another important consideration is whether smaller hospitals and clinics would actually be able to use these computational tools. Many of the current approaches require expensive equipment or specialized training that might not be available everywhere (12). For these tools to really make a difference in fighting antibiotic resistance, they need to be practical and accessible for all types of healthcare facilities, not just major medical centers with big research budgets.

Study Objectives

This literature review examines the current state of computational methods for predicting antimicrobial resistance in *K. pneumoniae*, specifically focusing on computer-based prediction approaches rather than traditional laboratory testing methods. By looking at which computational approaches show the most promise and identifying limitations that still need to be addressed, this research aims to help healthcare professionals determine when computational prediction tools might be helpful for treating patients. This analysis will also clarify situations where traditional testing methods remain necessary. The goal is to provide a clear picture of how these emerging computational technologies could improve patient care while avoiding the risks that come with relying too heavily on predictions that might not always be accurate (7),(8).

METHODS

Data Collection

The dataset used in this study was obtained from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) database, a bioinformatics resource that provides both genomic and phenotypic data for a wide range of bacterial pathogens (13). This database was chosen because it integrates high-quality, curated information from multiple public sources and includes standardized antimicrobial susceptibility testing (AST) results linked to bacterial genome assemblies. For *Klebsiella pneumoniae*, data were downloaded in two parts: the AST phenotype data, which show whether each bacterial isolate is resistant or susceptible to specific antibiotics, and the genome assembly files, which contain the full DNA sequence information for each isolate. Before analysis, all samples were filtered to remove entries with missing or incomplete AST results. AMR phenotype was then binarized, meaning that intermediate results were grouped with resistant ones to create a simple “resistant” versus “susceptible” outcome. This approach follows the standard practice in clinical microbiology, where intermediate isolates are treated as resistant to avoid underestimating resistance levels (7).

Resistance Gene Identification

AMR genes were identified using the ResFinder tool, which compares bacterial genome sequences against a curated database of known resistance genes (14), (15). ResFinder works by aligning the DNA sequences from each isolate to reference resistance genes using sequence homology searches. When a match above the default identity threshold (90%) is found, the gene is recorded as “present.” This method allows detection of both plasmid-encoded and chromosomal resistance genes, giving a comprehensive overview of the genetic basis of resistance in each isolate. The tool also provides rule-based AMR phenotype predictions by linking the presence of certain genes to resistance against specific antibiotics. To evaluate how well these genetic predictions matched real-world laboratory results, the ResFinder predictions were compared to the binarized phenotypic AST results for each antibiotic tested. The comparison was assessed using three common performance metrics: sensitivity, specificity, and accuracy (16). Sensitivity measures how well the model identifies resistant isolates, specificity measures how well it identifies susceptible ones, and accuracy summarizes overall correctness. All data cleaning, feature generation, and statistical analyses were conducted in Python using widely used scientific libraries such as Pandas for data manipulation and Scikit-learn for model evaluation. Custom scripts were written to organize gene presence/absence matrices, merge them with AST results, and calculate performance metrics. This computational approach ensured consistent, reproducible analysis across all antibiotics and isolates.

Machine Learning Analysis

To further assess the utility of the rule-based predictions, we sought to compare their performance to a machine learning (ML)-based prediction, which does not rely on rules but instead infers a best-fit model from a set of training samples. Machine learning models were developed using XGBoost, a gradient boosting algorithm that builds sequential decision trees to improve prediction accuracy (11). The presence or absence of each resistance gene detected by ResFinder was encoded as binary features (1 for present, 0 for absent), creating a feature matrix for each antibiotic. The features were then subsetted, so markers only known to confer resistance to the antibiotic of interest, as defined by ResFinder, were used to predict phenotype. A 5-fold stratified cross-validation approach was used to train and evaluate the models, ensuring balanced representation of resistant and susceptible isolates in each fold. XGBoost hyperparameters were set to max_depth=3 and n_estimators=30 to prevent overfitting while maintaining model interpretability.

Feature Importance Analysis

To understand which resistance genes contributed most to predictions for each antibiotic, SHAP (SHapley Additive exPlanations) values were calculated (17). SHAP values provide a unified measure of feature importance by quantifying how much each gene contributes to the model's prediction for individual samples. These values are based on game theory principles and offer transparent insight into which genetic markers drive resistance predictions for specific antibiotics.

RESULTS

Antimicrobial Resistance Patterns Across Antibiotics

Antibiotic	Total Samples	Resistant	Intermediate	Susceptible	Resistance Rate(%)
ampicillin	3,547	3,460	80	7	99.8
cefuroxime	2,801	2,655	18	128	95.4
ampicillin/sulbactam	2,143	1,763	123	257	88
ceftazidime	3,276	2,683	77	516	84.2
ceftriaxone	2,620	2,139	64	417	84.1
cefazolin	2,501	2,074	0	427	82.9
aztreonam	2,880	2,322	64	494	82.8
levofloxacin	2,123	1,637	53	433	79.6
ciprofloxacin	3,328	2,489	115	724	78.2
trimethoprim/sulfamethoxazole	2,960	2,237	15	708	76.1
piperacillin/tazobactam	2,899	1,755	282	862	70.3
ertapenem	1,850	1,249	11	590	68.1
cefepime	2,620	1,522	247	851	67
tobramycin	2,690	1,344	392	954	64.4
cefoxitin	2,996	1,603	239	1154	61.5
tetracycline	2,002	957	198	847	57.7
gentamicin	3,829	1,621	143	2065	46.1
meropenem	3,727	1,555	158	2014	46
imipenem	2,391	1,016	54	1321	44.8
amikacin	3,629	543	470	2616	27.9

Table 1: Resistance rates across antibiotics demonstrate dramatic variation across different antibiotic types.

The analysis of 4,003 *K. pneumoniae* isolates from the BV-BRC database showed major differences in antimicrobial resistance patterns across 20 different antibiotics tested (Table 1). Sample sizes varied significantly, ranging from 1,850 isolates tested for ertapenem to 3,829 isolates tested for gentamicin, which reflects how different antibiotics are used in hospitals. Ampicillin had the highest resistance rate at 99.8%, with 3,460 of 3,547 isolates being resistant and only 7 remaining susceptible. *K. pneumoniae* exhibits intrinsic resistance to ampicillin due to the chromosomally encoded SHV-1 β-lactamase that is constitutively expressed in this species (18). In contrast, amikacin had the lowest resistance rate at 27.9%, with 2,616 of 3,629 isolates remaining susceptible to treatment.

K. pneumoniae showed consistently high resistance to beta-lactam antibiotics. Cefuroxime had a 95.4% resistance rate (2,655 resistant out of 2,801 tested), while ampicillin/sulbactam reached 88.0% resistance. Third-generation cephalosporins also showed high resistance, with ceftazidime at 84.2% and ceftriaxone at 84.1%. Meropenem resistance reached 46.0%, while imipenem showed 44.8% resistance. *K. pneumoniae* isolates demonstrated high resistance to fluorquinolones, with 79.6% resistant to levofloxacin and 78.2% resistant to ciprofloxacin.

Distribution of Resistance Genes

Resistance Gene	Number of Isolates	Gene Function
oqxA	**3,649**	**Multidrug efflux pump**
oqxB	**3,644**	**Multidrug efflux pump**
fosA6	**2,521**	**Fosfomycin resistance**
catB3	**2,086**	**Chloramphenicol resistance**
sul1	**1,986**	**Sulfonamide resistance**
blaCTX-M-15	**1,894**	**Extended-spectrum beta-lactamase**
aph(3'')-Ib	**1,784**	**Aminoglycoside resistance**
sul2	**1,458**	**Sulfonamide resistance**

aph(6)-Id	**1,427**	**Aminoglycoside resistance**
blaTEM-1B	**1,317**	**Beta-lactamase**

Table 2: The most common resistance genes identified by ResFinder.

The genetic analysis revealed a wide variety of resistance genes across the *K. pneumoniae* isolates (Table 2). The efflux pump genes oqxA and oqxB were among the most common, detected in over 90% of isolates.

Beta-lactamase genes such as blaCTX-M-15 and blaTEM-1B were also frequently identified, which is consistent with the high resistance rates observed for beta-lactam antibiotics in Table 1. Similarly, sulfonamide resistance genes (sul1 and sul2) and aminoglycoside-modifying enzyme genes (aph(3")-Ib and aph(6)- Id) were common across isolates.

Performance of Rule-Based Predictions

Antibiotic	Total Samples	True Positives	False Positives	True Negatives	False Negatives	Sensitivity	Specificity	Accuracy
Ampicillin	**11786**	**11378**	**5**	**6**	**397**	**0.966**	**0.545**	**0.966**
Ceftazidime	**5870**	**5028**	**288**	**294**	**260**	**0.951**	**0.505**	**0.907**
Imipenem	**2425**	**924**	**82**	**1244**	**175**	**0.841**	**0.938**	**0.894**
Meropenem	**3881**	**1575**	**173**	**1848**	**285**	**0.847**	**0.914**	**0.882**
Gentamicin	**4004**	**1590**	**137**	**1929**	**348**	**0.82**	**0.934**	**0.879**
Aztreonam	**4886**	**4023**	**408**	**212**	**243**	**0.943**	**0.342**	**0.867**
Tobramycin	**3470**	**2230**	**260**	**717**	**263**	**0.895**	**0.734**	**0.849**
Ceftriaxone	**3921**	**3061**	**173**	**269**	**418**	**0.88**	**0.609**	**0.849**
Ertapenem	**1970**	**1016**	**4**	**586**	**364**	**0.736**	**0.993**	**0.813**
Ciprofloxacin	**8344**	**6594**	**1510**	**65**	**175**	**0.974**	**0.041**	**0.798**

Cefepime	**5143**	**3646**	**997**	**337**	**163**	**0.957**	**0.253**	**0.774**
Cefoxitin	**3124**	**1206**	**59**	**1097**	**762**	**0.613**	**0.949**	**0.737**
Tetracycline	**2049**	**695**	**48**	**800**	**506**	**0.579**	**0.943**	**0.73**
Amikacin	**4084**	**1215**	**1153**	**1533**	**183**	**0.869**	**0.571**	**0.673**

Table 3: Performance Metrics for Rule-Based ResFinder Predictions Across 14 Antibiotics

To evaluate how well ResFinder's rule-based gene presence approach predicts antibiotic resistance, performance metrics were calculated for each antibiotic across all available isolates (Table 3). Across all antibiotics, ResFinder demonstrated generally high sensitivity, often above 0.80, indicating that it successfully identified most resistant isolates in many cases. For example, ampicillin (0.966), ceftazidime (0.951), cefepime (0.957), ciprofloxacin (0.974), aztreonam (0.943), and tobramycin (0.895) all showed sensitivities above 0.80, meaning the majority of resistant isolates were correctly classified. This indicates that the rule-based method is effective at detecting the presence of known resistance genes associated with phenotypic resistance. However, specificity values were considerably lower and more variable, ranging from near zero for ciprofloxacin (0.041) to above 0.90 for ertapenem (0.993), imipenem (0.938), cefoxitin (0.949), tetracycline (0.943), and gentamicin (0.934). This indicates that while the model rarely misses resistant strains for most antibiotics, it frequently misclassifies susceptible isolates as resistant for others. For example, ciprofloxacin, cefepime (0.253), and aztreonam (0.342) showed very high false positive rates, with 1,510, 997, and 408 false positives respectively, reflecting the challenge of distinguishing between gene presence and actual phenotypic resistance.

Overall accuracy ranged from 0.673 to 0.966, with the highest performance seen for ampicillin (0.966), ceftazidime (0.907), imipenem (0.894), and meropenem (0.882). In contrast, amikacin showed the lowest accuracy (0.673) despite reasonable sensitivity (0.869), largely due to poor specificity (0.571) resulting in 1,153 false positives. The wide variability in accuracy highlights how the reliability of genetic resistance predictions depends heavily on the antibiotic and its underlying resistance mechanisms. These findings underscore the limitations of purely rule-based prediction systems.

Machine Learning Model Performance

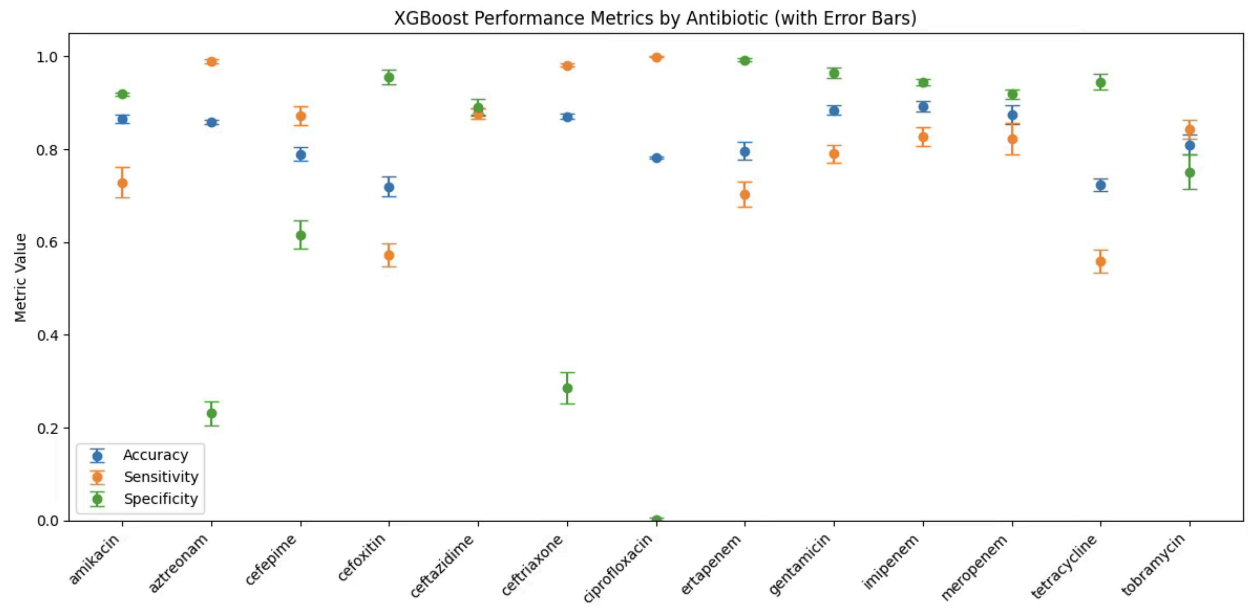

Figure 1: XGBoost model performance metrics for predicting antimicrobial resistance across antibiotics.

Figure 1 shows 13 antibiotics (excluding ampicillin which was removed due to its 99.8% resistance rate making meaningful model comparison impossible). Seven antibiotics from the full dataset do not appear in this comparison: ampicillin/sulbactam, cefazolin, cefuroxime, levofloxacin, piperacillin/tazobactam, and trimethoprim/sulfamethoxazole lack sufficient ResFinder annotations for XGBoost training, limiting the analysis to antibiotics where both rule-based and machine learning predictions could be generated.

To evaluate whether machine learning could improve upon simple rule-based gene presence predictions, XGBoost models were trained and compared against ResFinder's rule-based approach across 14 antibiotics (Figure 4). Ampicillin was excluded from this comparison because its resistance rate was nearly universal (99.8%), leaving too few susceptible samples for effective model training. Overall, the XGBoost models achieved high accuracy for most antibiotics, often exceeding 0.85, which indicates strong agreement between predicted and actual phenotypes. Sensitivity values were generally high as well, showing that the models could correctly identify most resistant isolates. However, specificity varied more widely across antibiotics, reflecting differences in how well the model distinguished susceptible isolates from resistant ones. Antibiotics such as aztreonam, ceftriaxone, and ceftazidime showed both high sensitivity and specificity. Certain antibiotics like cefepime and ciprofloxacin showed lower specificity values. XGBoost models achieved high accuracy for most antibiotics, often exceeding 0.85.

Comparison of Prediction Approaches

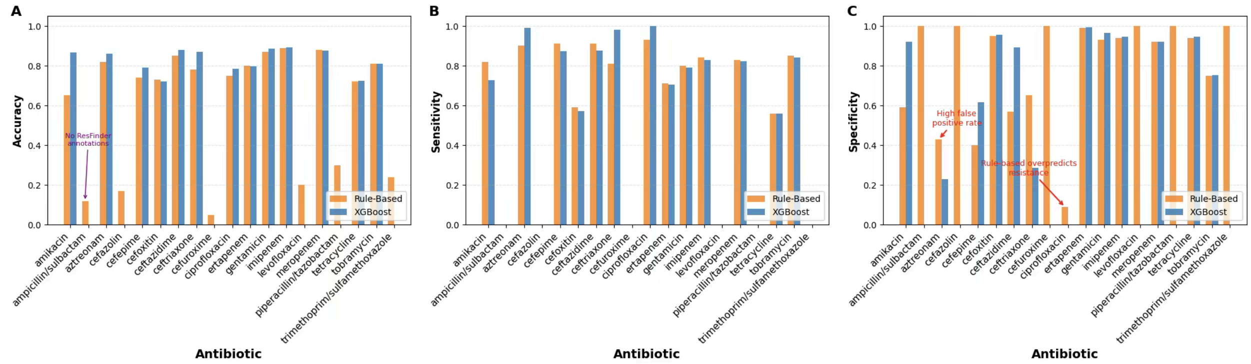

Figure 2: Comparison of rule-based ResFinder and XGBoost model performance across all antibiotics by A) accuracy, B) sensitivity and C) specificity.

Both the rule- and ML-based approaches achieved high sensitivity across most antibiotics, indicating strong ability to detect resistant isolates (Panel B). The rule-based ResFinder method showed sensitivity ranging from approximately 0.56 (tetracycline) to 0.99 (aztreonam, ertapenem), while XGBoost models achieved sensitivity between 0.56 (tetracycline) and 1.0 (aztreonam, ertapenem, ciprofloxacin). However, specificity, the ability to correctly identify susceptible isolates, varied considerably between methods and across antibiotics (Panel C). For several antibiotics, XGBoost demonstrated substantially improved specificity compared to the rule-based approach. The most notable improvements were observed for amikacin (0.59 rule-based vs 0.92 XGBoost), ceftazidime (0.57 vs 0.89), and cefepime (0.40 vs 0.62).

Interestingly, some antibiotics showed decreased specificity with XGBoost compared to rule-based methods, including ceftriaxone (0.65 rule-based vs 0.29 XGBoost), aztreonam (0.43 vs 0.23), and ciprofloxacin (0.09 vs 0.003). However, some antibiotics showed only modest changes in specificity. For tetracycline, both methods achieved high specificity (approximately 0.94 for both), while for imipenem, meropenem, gentamicin, and ertapenem, both approaches achieved similarly high specificity (>0.90).

Overall accuracy favored XGBoost for most antibiotics (Panel A), with the largest improvements seen for amikacin (0.65 rule-based vs 0.87 XGBoost), ceftriaxone (0.78 vs 0.87), cefepime (0.74 vs 0.79), aztreonam (0.82 vs 0.86), ceftazidime (0.85 vs 0.88), and ciprofloxacin (0.75 vs 0.78). Notably, antibiotics where both methods performed similarly well in specificity also showed similar accuracy values, such as imipenem (0.89 rule-based vs 0.89 XGBoost) and meropenem (0.88 vs 0.88). The substantial accuracy improvement for amikacin (21.6 percentage points) demonstrates the potential value of machine learning for antibiotics where rule-based predictions struggle with high false positive rates.

Feature Importance Analysis

SHAP analysis revealed which resistance genes were most influential for predicting resistance to specific antibiotics. For amikacin resistance, the most important genes were armA and aac(6')-Ib, both aminoglycoside-modifying enzymes known to confer high-level amikacin resistance through enzymatic modification of the drug (22). The rmtB gene, encoding a 16S rRNA methyltransferase, also contributed substantially, consistent with its known role in aminoglycoside resistance. For ciprofloxacin, the analysis identified qnr genes (plasmidmediated quinolone resistance) and mutations in gyrA and parC as key predictors, while efflux pump genes like oqxA and oqxB showed moderate importance. Beta-lactam antibiotics showed expected patterns, with blaCTX-M variants being highly important for ceftriaxone and ceftazidime predictions, while carbapenemase genes (blaKPC, blaNDM, blaOXA-48) dominated predictions for meropenem and imipenem resistance. Interestingly, some genes showed cross-antibiotic importance. The efflux pump genes oqxA and oqxB contributed to predictions across multiple drug classes, reflecting their multidrug efflux capability. This pattern highlights how machine learning models can capture pleiotropy, where single genes affect resistance to multiple antibiotics. The SHAP analysis also revealed that for some antibiotics, no single gene was overwhelmingly predictive. Instead, resistance appeared to be determined by combinations of multiple genes with moderate individual effects.

DISCUSSION

Clinical Significance of Resistance Patterns

The findings of this study highlight the widespread prevalence of resistance genes and phenotypic resistance in *K. pneumoniae*. The high frequency of efflux pump genes (OqxA, OqxB) underscores the bacterium's ability to resist multiple antibiotic classes simultaneously (3), (4). The presence of blaCTX-M-15 aligns with global reports of ESBL-producing *K. pneumoniae*, which severely limit treatment options (23), (21). Phenotypic results further confirmed the clinical challenge showing nearly universal ampicillin resistance and high rates of resistance to cephalosporins and carbapenems making many firstline and some last-line therapies ineffective. While aminoglycosides and tetracycline retain partial activity, their effectiveness is limited by toxicity concerns and rapid resistance development (2).

Genetic Basis of Resistance

The widespread distribution of resistance genes show important insights into the relationship between genotype and phenotype. The efflux pump genes oqxA and oqxB were found in over 90% of isolates, but still their presence in both resistant and susceptible samples indicates that gene presence alone does not always guarantee functional resistance. This could be due to differences in gene expression levels, regulatory mechanisms, or the influence of other resistance pathways that determine whether the gene is actively contributing to resistance in a given isolate. Interestingly, although these genes were present in both resistant and susceptible samples, their varying combinations and interactions likely influenced the actual resistance phenotype. The chloramphenicol resistance gene catB3 was also widely distributed, further illustrating the genetic diversity of resistance mechanisms within this bacterial population. Overall, the genetic analysis demonstrates that *K. pneumoniae* carries a diverse set of resistance genes, but gene detection alone is not always enough to predict actual resistance. This highlights the complexity of the genetic and environmental factors that determine antimicrobial resistance expression.

The moderate carbapenem resistance rates (46.0% for meropenem, 44.8% for imipenem) are concerning given the fact that carbapenems are often used as last resort antibiotics when other treatments fail. The emergence of carbapenem resistance severely limits therapeutic options and poses significant challenges for infection control in healthcare settings.

Comparison of Computational Approaches

The comparison between rule-based and machine learning approaches revealed important trade-offs in computational AMR prediction. While ResFinder's gene presence method achieved high sensitivity, the poor specificity for many antibiotics (particularly fluoroquinolones and some beta-lactams) would result in substantial overprediction of resistance in clinical practice. In real-world settings, this overprediction could lead clinicians to unnecessarily escalate to broad-spectrum antibiotics like carbapenems or colistin when narrower-spectrum agents would have been effective. This not only drives further resistance development through selective pressure but also exposes patients to increased risks of toxicity and adverse effects associated with more aggressive therapies. Furthermore, in resource-limited settings where access to newer or last-line antibiotics is restricted, overpredicting resistance could leave clinicians with fewer perceived treatment options, potentially delaying appropriate care.

XGBoost models substantially improved specificity while maintaining high sensitivity, suggesting that considering gene combinations and interactions provides more accurate resistance predictions than simple gene presence. The quality and completeness of genomic data significantly influence the performance gap between these two approaches. Rule-based methods like ResFinder depend on well curated databases of known resistance genes and established genotype-phenotype associations. When resistance mechanisms are well-characterized and consistent across isolates, rule-based predictions perform adequately. However, machine learning approaches like XGBoost can leverage patterns in incomplete or noisy data by learning complex relationships between multiple genetic features. This becomes particularly valuable when dealing with real-world genomic data that may have sequencing errors, incomplete gene annotations, or novel resistance mechanisms not yet cataloged in reference databases. The ability of XGBoost to integrate information from multiple genes simultaneously allows it to compensate for uncertainties in individual gene calls and to detect resistance patterns that emerge from combinations of genes rather than single determinants.

Clinical Implications of Prediction Accuracy

The SHAP analysis provided clinically valuable insights into which resistance mechanisms drive predictions for each antibiotic. The clear identification of armA and aac(6')-Ib as primary amikacin resistance determinants, for example, could guide targeted molecular diagnostics. Similarly, understanding that ciprofloxacin resistance predictions rely on combinations of chromosomal mutations and plasmid-mediated genes helps explain why simple gene presence may be insufficient for this antibiotic. Fluoroquinolone resistance in *K. pneumoniae* typically arises through stepwise accumulation of mutations in DNA gyrase (gyrA) and topoisomerase IV (parC) genes, with each mutation conferring incremental increases in resistance levels (25). Clinically relevant chromosomally encoded multidrug resistance efflux pumps in bacteria. Clinical Microbiology Reviews, 19(2), 382-402. https://doi.org/10.1128/CMR.19.2.382-402.2006)). The presence of plasmid-mediated quinolone resistance (PMQR) genes such as qnrA, qnrB, and qnrS can further elevate resistance when combined with chromosomal mutations, creating a complex multi-factorial resistance phenotype that cannot be captured by detecting single genes alone (26). Beyond these examples, the SHAP analysis revealed interesting patterns for other antibiotics as well. For the carbapenems imipenem and meropenem, carbapenemase genes (blaKPC, blaNDM, blaOXA-48) dominated predictions, which aligns with their known clinical importance and suggests that targeted PCR screening for these genes could remain highly effective. In contrast, for ceftazidime and ceftriaxone, multiple blaCTX-M variants contributed to predictions along with other ESBL genes, indicating that resistance to these third generation cephalosporins arises from a more diverse genetic basis. Tobramycin predictions showed contributions from multiple aminoglycoside-modifying enzyme genes, reflecting the overlapping substrate specificities of these enzymes. For some antibiotics like tetracycline, the relatively modest difference between rule-based and machine learning performance suggests that resistance mechanisms are simpler and better captured by gene presence alone, whereas the dramatic improvements seen for fluoroquinolones indicate that their resistance involves more complex genetic interactions that benefit from machine learning approaches.

Limitations

The diagnostic evaluation of computational prediction tools revealed important limitations. Although sensitivity was relatively high, the low specificity indicates a tendency to overpredict resistance. This aligns with prior studies noting that genotypic methods may identify resistance genes that are not always expressed or clinically relevant (10). Such discrepancies underscore the complexity of correlating genotype with phenotype, especially in bacteria with diverse resistance mechanisms (24), (13). Nevertheless, computational methods remain promising as rapid screening tools. Their ability to detect resistance-associated genes could guide initial treatment decisions while awaiting confirmatory phenotypic results. Integrating machine learning approaches that account for gene expression, resistance gene context, and environmental factors may improve predictive accuracy (7), (8).

However, several limitations warrant consideration. The machine learning models require sufficient training data, which may not be available for all antibiotics and would not encompass all geographic regions. Geographic variation in resistance mechanisms poses a significant challenge for predictive models. Some countries, particularly in low- and middle-income regions, are substantially underrepresented in public genomic databases due to limited sequencing infrastructure and resources. This means that models trained primarily on data from high-income countries may fail to capture resistance mechanisms or genetic variants that are prevalent in other regions. For example, certain carbapenemase variants or ESBL types may be endemic in specific geographic areas but rare in the databases used for model training. Studies have documented substantial geographic variation in carbapenemase distribution, with blaKPC predominating in North America and parts of Europe, blaNDM more common in South Asia and the Middle East, and blaOXA-48 prevalent in Mediterranean countries and North Africa (27), (28). Similarly, ESBL variants show geographic clustering, with blaCTX-M-15 widespread globally but other CTX-M types showing regional specificity (29). Additionally, local antibiotic usage patterns, prescribing practices, and infection control measures create different selective pressures that shape the evolution of resistance mechanisms. A model trained on European isolates might not accurately predict resistance patterns in Southeast Asian or Sub-Saharan African isolates if the underlying resistance gene repertoires or their prevalence differ substantially. This geographic bias in training data could lead to systematic underprediction or overprediction of resistance in underrepresented populations, potentially exacerbating health disparities.

The models also assume that the relationship between genotype and phenotype remains stable, which may not hold as new resistance mechanisms emerge or as bacterial populations evolve. Horizontal gene transfer through plasmids, transposons, and integrons allows resistance genes to spread rapidly between different bacterial species and strains, sometimes crossing genus boundaries. This mobility means that resistance genes observed in one bacterial context during model training may later appear in entirely different genetic backgrounds where their expression levels, regulatory control, or phenotypic effects differ. For example, *K. pneumoniae* has been described to have a highly dynamic genome, readily acquiring resistance determinants from environmental reservoirs and other bacterial species through horizontal gene transfer (2), (30). Furthermore, bacteria can acquire novel resistance mechanisms through horizontal transfer of gene cassettes from environmental reservoirs that were not represented in the training data at all. The dynamic nature of bacterial genomes, with resistance genes frequently moving between chromosomal and plasmid locations, means that the specific genetic contexts that influence gene expression and resistance phenotypes are constantly shifting. Machine learning models that treat gene presence as static features may fail to account for these evolutionary dynamics, leading to prediction errors when resistance determinants appear in new combinations or genetic contexts not seen during training.

Implementation Challenges in Resource-Limited Settings

The practical implementation of computational AMR prediction tools in low-resource settings presents both opportunities and challenges. On one hand, these tools could be transformative in regions where traditional culture-based AST is unavailable, slow, or prohibitively expensive. Rapid molecular diagnostics paired with computational prediction could enable targeted antibiotic therapy without the need for extensive laboratory infrastructure. However, several barriers limit immediate applicability. First, even simplified sequencing approaches like targeted amplicon sequencing or PCR-based resistance gene detection require equipment, reagents, and trained personnel that may not be consistently available in resource-limited healthcare facilities. Second, as discussed above, the underrepresentation of local resistance mechanisms in training databases could lead to poor predictive performance precisely in the settings where these tools are most needed. Third, clinical decision support systems need to be adapted to local antibiotic availability and treatment guidelines, as the antibiotics accessible in a rural clinic in sub-Saharan Africa may differ substantially from those in a tertiary hospital in North America or Europe. For computational AMR prediction to achieve meaningful global health impact, efforts must focus on expanding genomic surveillance in underrepresented regions, developing low-cost and robust molecular diagnostic platforms suitable for field deployment, and creating region-specific prediction models that account for local resistance epidemiology. Additionally, any implementation must be accompanied by training programs for healthcare workers and integration into existing clinical workflows to ensure that predictions are interpreted appropriately and used to guide rather than replace clinical judgment.

CONCLUSION

This study demonstrates that computational approaches for predicting antimicrobial resistance in *K. pneumoniae* show potential but are not yet sufficient to replace traditional laboratory testing. The widespread presence of multidrug resistance genes and high phenotypic resistance rates confirms the urgent threat posed by this pathogen. Computational predictions achieved strong sensitivity but poor specificity, highlighting their limitations in accurately distinguishing resistant from susceptible strains. While ResFinder could identify most resistant bacteria, the high number of false positives means doctors would still need to be cautious about relying on these predictions alone. The challenge of translating genetic information into clinical decisions became clear when analyzing the performance metrics - having resistance genes doesn't always mean the bacteria will actually be resistant in real infections. Future research should focus on improving the specificity of computational tools by incorporating additional biological context, such as gene expression levels and resistance gene interactions. The same resistance gene might behave differently depending on the bacterial strain, the patient's condition, and even the specific laboratory conditions used for testing. Expanding datasets to include diverse global isolates and developing accessible tools for resource-limited settings will be essential steps toward making computational prediction a practical asset in the fight against antimicrobial resistance. This research has shown that while computational methods are promising, the gap between laboratory predictions and real-world clinical utility remains significant and will require continued innovation to bridge effectively.

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